HOLY ANGEL UNIVERSITY

# IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science

Quebec Student
Romeo Student
Juliett Student
X-ray Student

April 2026

*Proposal manuscript — revised after the proposal defense (see Summary of Revisions).*
*Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are placeholders.*

SCHOOL OF COMPUTING

## APPROVAL SHEET

This capstone entitled **"IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science has been examined and is recommended for acceptance and oral examination.

**Dean Delta**
Adviser

## ORAL EXAMINATION

Approved by the committee of oral examiners on
April 13, 2026

**Prof. Foxtrot**
Panel Chair

**Prof. Bravo**
Panel Member

**[Name of Panel Member]**
Panel Member

## APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science.

**Prof. Alpha**
Program Chair/Coordinator,
Computer Science

**Dean Delta**
Dean, School of Computing

**Acknowledgment**

The researchers extend their sincere gratitude to everyone who supported this capstone project. This page is placeholder text: a real manuscript thanks the Adviser, the panel, the instructors, the host office and the families who made the work possible.

The group is especially grateful to Dean Delta for patient guidance through every revision.

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**Summary of Revisions**

The following revisions were made in response to the panel's recommendations recorded in the Capstone Revision Form (FM-AAC-SOC-2004) after the proposal defense on April 13, 2026.

| Panel recommendation | Action taken | Section |
|---|---|---|
| Tighten the scope statement: say which services are in and which are out. | Scope and Delimitations now lists the covered services and the exclusions in two paragraphs. | Scope and Delimitations |
| Make objective 3 measurable. | Objective 3 now states the measured targets from the non-functional requirements. | Objectives of the Study |
| Add the respondents' selection criteria. | Sample and Setting now explains the purposive sampling criteria for each group. | Sample and Setting |

Checked by: _____ Dean Delta, Adviser

**IoT-Based Laboratory Equipment Tracking for Computing Laboratories**

Quebec Student, Romeo Student, Juliett Student, X-ray Student

Holy Angel University
Computer Science

**Abstract**

*The abstract is written after the study is completed and is left blank at the proposal stage.*

# IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

## Introduction

The computing laboratories lend projectors, development boards, networking kits and spare peripherals to classes every day. Borrowing is recorded on paper, items are moved between rooms without a record, and the semestral inventory regularly finds equipment missing or in the wrong room.

Bluetooth Low Energy (BLE) tags are inexpensive and last months on a coin cell. Gateways in each room can detect the tags and report their signal strength, from which a server can estimate the room an item is in. Combined with a borrowing log, such a system would show where every item is and flag items that leave the laboratories.

The problem this study addresses is the following: laboratory staff cannot locate equipment without searching every room, borrowing records are incomplete, and losses are discovered only at inventory time. This capstone project therefore proposes the equipment tracker.

## Review of Related Literature

This section reviews studies and existing systems related to the equipment tracker and the methods used to evaluate it.

### *Indoor Positioning with Bluetooth Low Energy*

RSSI-based room-level positioning with one receiver per room achieves over 90% accuracy when readings are smoothed over a short window (Placeholder & Author, 2023).

### *Asset Tracking in Educational Laboratories*

Tagged asset systems in school laboratories reduced search time and made borrowing records complete (Sample, 2024).

### *Signal Smoothing and Classification*

Moving averages and Kalman filters reduce RSSI noise; a nearest-gateway rule with hysteresis prevents items from flickering between rooms (Example & Writer, 2022).

### *Evaluating IoT Systems*

ISO/IEC 25010 is commonly applied to IoT prototypes alongside accuracy measurements of the sensing layer (Example, 2023).

### *Synthesis*

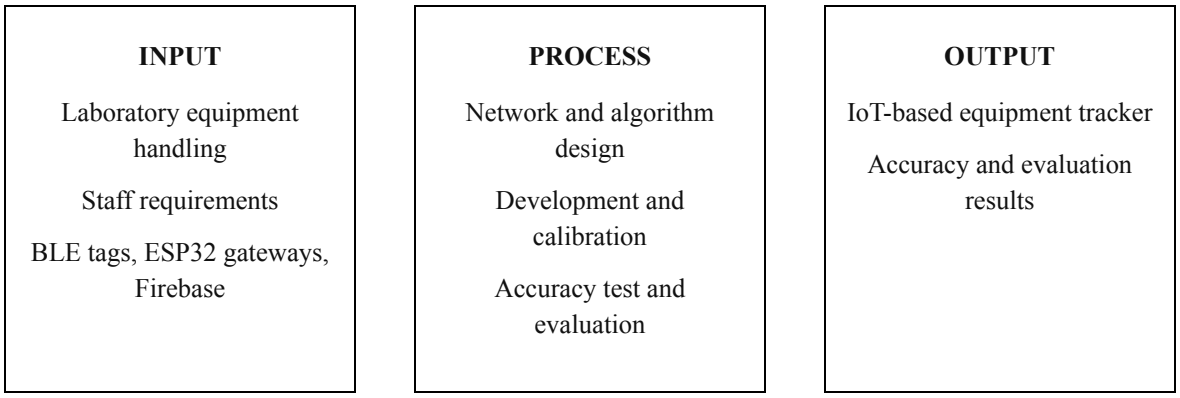
The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the equipment tracker fills.

## Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

**Figure 1**

*Conceptual Framework of the Study*




## Objectives of the Study

The general objective of the study is to design, build and evaluate an IoT system that locates tagged laboratory equipment by room and records its movement. Specifically, the study aims to:

1. determine the equipment-handling problems of the computing laboratories through observation and interviews;
2. design a BLE tag and gateway network and a room-level location algorithm based on received signal strength;
3. develop a web dashboard that shows the location and movement history of each item and alerts staff when an item leaves the laboratories;
4. measure the room-level location accuracy of the algorithm in the four computing laboratories; and
5. evaluate the system using ISO/IEC 25010 with laboratory staff and IT experts.

## Scope and Delimitations

The study covers the four computing laboratories and the equipment room on one floor, using BLE tags on 60 items and one gateway per room. Location is resolved to the room, not to a position within it.

The study does not track people, does not cover equipment outside the floor, and does not replace the university's property inventory system; it provides location and movement records that staff can reconcile with it.

## Method

This section describes how the equipment tracker will be designed, built, measured and evaluated.

### *Research Design*

The study uses a developmental research design with a quantitative evaluation. The location algorithm is developed iteratively and its accuracy is measured experimentally.

**Table 1**

*Functional Requirements*

| ID | User | Requirement |
|---|---|---|

| | | |
|---|---|---|
| FR-01 | Staff | Register an item and attach a BLE tag |
| FR-02 | System | Estimate the room of each tagged item every five seconds |
| FR-03 | Staff | View the location and movement history of an item |
| FR-04 | System | Alert staff when an item leaves the laboratories |
| FR-05 | Staff | Record borrowing and return of an item |
| FR-06 | Supervisor | Generate inventory and movement reports |

**Table 2**

*Non-Functional Requirements*

| Characteristic | Requirement |
|---|---|
| Accuracy | Room-level location is correct for at least 90% of moves |
| Performance | A move appears on the dashboard within ten seconds |
| Reliability | A gateway that goes offline is reported within one minute |
| Security | Only laboratory staff accounts can view locations |

***Sources of Data***

RSSI readings from 60 tagged items collected by four gateways every second for two weeks, and ground-truth room labels recorded by the researchers during 400 controlled moves.

***Participants***

Five laboratory staff and five IT experts evaluate the system; no personal data about students is collected.

**Table 3**

*Respondents of the Study*

| Respondent group | Number | Instrument |
|---|---|---|
| Laboratory staff | 5 | ISO/IEC 25010 |
| IT experts | 5 | ISO/IEC 25010 |

***Instruments***

The ISO/IEC 25010 evaluation form (Appendix H) and an accuracy test log recording the true and predicted room of each move.

***Data Collection***

After approval from the Laboratory Supervisor, the researchers install the gateways, tag the items, and collect readings for two weeks while recording controlled moves.

***Data Analysis***

Accuracy is the share of moves whose predicted room matches the true room after the smoothing window. ISO/IEC 25010 ratings are summarized by weighted mean (Table 6).

***Ethical Consideration***

Participation is voluntary and respondents may withdraw at any time. Responses are anonymous, kept for the duration of the study and destroyed after the final defense, in accordance with the Data Privacy Act of 2012.

## References

Author, A. B., Writer, C. D., & Name, E. F. (2023). A placeholder study of progressive web applications for campus services. Journal of Sample Computing, 12(3), 45–58.

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Sample, O. (2024). Asset tracking in teaching laboratories. Sample Journal of Educational Technology, 2(2), 60–72.

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**Appendix A**

**Cover Letter to Host/Locale**

February 16, 2026

The Laboratory Supervisor
Computing Laboratories Office
Holy Angel University

Dear Sir:

We are students of the School of Computing working on our capstone project entitled “IoT-Based Laboratory Equipment Tracking for Computing Laboratories.” We respectfully ask permission to observe the current process at your office, interview your staff, and conduct a test session of the system. This letter is a placeholder for the dummy manuscript.

Respectfully yours,

Quebec Student
Romeo Student
Juliett Student
X-ray Student

Noted by:
Dean Delta, Adviser

## Appendix B

## Current Process Flowchart

Request is made by message or on paper

Staff checks availability manually

Request is confirmed or declined

Activity takes place

Record is kept on paper, if at all

## Appendix C

### Functionality Checklist

| Function | Working | Not working | Remarks |
|---|---|---|---|
| Register item and tag | | | |
| Room estimate every five seconds | | | |
| Location history | | | |
| Out-of-laboratory alert | | | |
| Borrowing and return | | | |
| Inventory report | | | |
| Gateway offline alert | | | |

## Appendix D

### System Usability Scale Questionnaire

1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 – Strongly Agree

| Statement | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1. I think that I would like to use this system frequently. | | | | | |
| 2. I found the system unnecessarily complex. | | | | | |
| 3. I thought the system was easy to use. | | | | | |
| 4. I think that I would need the support of a technical person to be able to use this system. | | | | | |
| 5. I found the various functions in this system were well integrated. | | | | | |
| 6. I thought there was too much inconsistency in this system. | | | | | |
| 7. I would imagine that most people would learn to use this system very quickly. | | | | | |
| 8. I found the system very cumbersome to use. | | | | | |
| 9. I felt very confident using the system. | | | | | |
| 10. I needed to learn a lot of things before I could get going with this system. | | | | | |

## Appendix E

## ISO/IEC 25010 Software Quality Evaluation Form

| Criteria | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| **Functional Suitability** | | | | | |
| The system meets the functional suitability requirements. | | | | | |
| **Performance Efficiency** | | | | | |
| The system meets the performance efficiency requirements. | | | | | |
| **Usability** | | | | | |
| The system meets the usability requirements. | | | | | |
| **Reliability** | | | | | |
| The system meets the reliability requirements. | | | | | |